

2. D) we need to show that X and X^{-1} cannot both be corrected. Let's use KL:

$$E_a = X \quad ; \quad E_b = X^{-1}$$

$$E_a^\dagger E_b = X^\dagger \cdot X^{-1} \quad \left[X^{-1} = X^7 \right]$$

$$\Rightarrow E_a^\dagger E_b = X^\dagger \cdot X^7 = X^6$$

$$\Rightarrow X^6 = X^4 \cdot X^2 = S_1 \cdot X_L = X_L$$

$$\therefore E_a^\dagger E_b = X_L \text{ (logical error)}$$

$$\langle 0_L | X_L | 0_L \rangle = \langle 0_L | 1_L \rangle = \langle 1_L | X_L | 0_L \rangle$$

$$\langle 0_L | X_L | 1_L \rangle = 1 = \langle 1_L | X_L | 0_L \rangle$$

$\therefore$ KL conditions fail.

... QED.